

Name of Program	Bachelor of Technology (Computer Science Engineering)					
CSET371	Big Data Analytics and Business Intelligence	L	T	P	C	
Owning School/Department	School of Computer Science Engineering and Technology	2	0	2	3	
Pre-requisites/Exposure	-					

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: To articulate key technologies in data science and business analytics.

CO2: To build Big Data framework: Hadoop (HDFS and MapReduce), Hadoop Ecosystem & spark.

CO3: To make use of cutting-edge tools and technologies to analyze Big Data.

CO-PO/PSO Mapping

	PO 1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO2	PSO3
CO1	3	2	3	2	3	2	3	2		3	2	2	2	3	2
CO2	2	2	3	2	2	2				2	2	1	3	3	2
CO3	3	1	1	1	2								2	2	1

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module I:

8 lecture hours

Big Data Analytics: Data and Relations, Business Intelligence, Business intelligence vs business analytics, Big Data Predictive Analysis, Why what and how BI?, OLTP VS OLAP, Project life cycle, Methodology, Impact Analysis, Testing and release management, Ethics in Business Intelligence, Big Data Technology Component, Real Time Analysis of Big Data, Big Data Architecture, Cost Conscious Design, Storage Conscious Design.

Modulo II:

6 lecture hours

Big Data Warehouse, Functional vs Procedural programming models, Data placement strategies, CAP Theorem, Streaming, Stream Data Types, Stream Computing, Filtering Stream, Estimating Moments, NoSQL, Aggregate Data Models, Document Data Model, Schema Less Databases.

Module III:

6 lecture hours

Hadoop Ecosystem, Hadoop Overview, HDFS, Map-Reduce, Pig Overview, Pig Grunt Shell, Hive, HBase, Architecture of Hive and HBase.

Module IV:

8 lecture hours

HQL, Associations and Joins, Aggregate function, Polymorphic queries, Clauses, Subqueries, Spark, Core, Spark SQL, Spark RDD, Deployment and Integration, Spark GraphX and Graph Analytics, Research Topics in Big Data Analytics.

Studio Work / Laboratory Experiments:

Students will gain practical experience by using tools and technologies like R, HDFS, MapReduce, Hive, pig, NoSQL, tableau.

Text Books :

1. Ghavami Peter , *Big Data Analytics Methods* (2nd ed.), De Gruyter, 2020. ISBN 9781547417951.
2. Acharya and Seema, *Data Analytics using R* (1st ed.), New York: McGraw-Hill Education, 2018. ISBN 9352605241.

Reference Books :

1. Azevedo Ana and Manuel Filipe Santos, *Integration Challenges for Analytics, Business Intelligence, and Data Mining* (1st ed.), Engineering Science Reference, 2020. ISBN 9781799857832.

Assessment Scheme:

Components	Internal Assessment	Mid Term Exam	End Exam	Total
Weightage (%)	45%	20%	35%	100%